

1. Course number: 010153002
Course name: Computer Programming
2. Credits: 3(2-2-5)
Contact hours: 3 hours per week
3. Instructor:
Asst. Prof. Dr. Ruslee Sutthaweeikul
4. Textbook:
 - a. Forouzan, Behrouz A., and Richard F. Gilberg. *C++ Programming: An Object-oriented Approach*. McGraw-Hill, 2020. ([*KMUTNB library*](#))
5. Specific course information:
 - a. Course Description: Basic operations of a computer, required components of a computer, how hardware and software work together, data processing, fundamental of high level computer programming, application design and development, problem solving by using computer programming.
 - b. Prerequisites: -
 - c. Course type: Required
6. Specific goals for the course (Course Learning Outcome: CLO)
 - a. Specific outcomes of instruction:
 1. Understand computer systems concepts, different computing environments, computer languages, and the system development life cycle.
 2. Write programs using basic programming constructs including identifiers, data types, variables, constants, input/output statements, logical and comparative operators, and control structures (selections and loops).
 3. Understand and apply expressions, precedence rules, associativity in evaluating expressions, type conversions, and various statement types.
 4. Develop and implement programs using arrays, pointers, and functions, including sorting algorithms, search algorithms, and string manipulations.
 5. Understand and utilize complex data structures such as structures, unions, and enumerated types.
 6. Recognize the importance of effective communication, ethical and professional responsibilities in engineering, and the impact of engineering solutions in various contexts.

b. Student Outcomes Addressed:

ABET Student Outcome (SO)	Course Learning Outcome: CLO
1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	1-6
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1-6
3. an ability to communicate effectively with a range of audiences	6
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	6
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	-
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1-6
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies	1-6

7. Course Learning Outcomes and Assessment tools:

Course Learning Outcome: CLO	SO	Assessment tools
1. Understand computer systems concepts, different computing environments, computer languages, and the system development life cycle.	1,7	Exams, Quiz, Assignment, Project
2. Write programs using basic programming constructs including identifiers, data types, variables, constants, input/output statements,	1,2,7	Exams, Quiz, Assignment, Project

Course Learning Outcome: CLO	SO	Assessment tools
logical and comparative operators, and control structures (selections and loops).		
3. Understand and apply expressions, precedence rules, associativity in evaluating expressions, type conversions, and various statement types.	1,2,7	Exams, Quiz, Assignment, Project
4. Develop and implement programs using arrays, pointers, and functions, including sorting algorithms, search algorithms, and string manipulations.	1,2,7	Exams, Quiz, Assignment, Project
5. Understand and utilize complex data structures such as structures, unions, and enumerated types.	1,7	Exams, Quiz, Assignment, Project
6. Recognize the importance of effective communication, ethical and professional responsibilities in engineering, and the impact of engineering solutions in various contexts.	1,7	Exams, Quiz, Assignment, Project

8. Grading Plan:

Score	Grade
80-100	A
75-79	B+
70-74	B
65-69	C+
60-64	C
55-59	D+
50-54	D
0-49	F

9. Brief list of topics to be covered:

- a. Introduction to Computer
- b. Introduction to Computer Language
- c. Structure of a Program

- d. Selection-Making Decisions
- e. Repetition
- f. Arrays
- g. Pointers
- h. Functions
- i. Strings
- j. Enumerated, Structure, and Union Types